

Week 3:

Day 2 – Your First Machine Learning Model

Today you will train your first ML model.

Learn

Study:

- Training data
- Testing data
- x and y
- `train_test_split()`
- Model training
- Predictions

Understand this basic process:

```
Dataset
↓
Prepare Data
↓
Split Data
↓
Train Model
↓
Make Predictions
↓
Evaluate Model
```

Practical Work

Use the Iris dataset.

Program 1 – Train a Classification Model

Use:

Decision Tree

Train a model to predict the type of Iris flower.

You should:

1. Load the dataset.
2. Separate features and labels.
3. Split the dataset into training and testing data.
4. Create a Decision Tree model.
5. Train it.
6. Make predictions.
7. Check the accuracy.

Program 2 – Make Your Own Prediction

Give the model new measurements and ask it to predict the flower type.

For example:

```
Sepal Length  
Sepal Width  
Petal Length  
Petal Width
```

The model should return a predicted class.

Important

Don't just copy the code.

You should be able to explain:

What does `fit()` do?

What does `predict()` do?

Why do we split the dataset?

Git Task

Commit and push today's work.

Day 3 – Evaluating a Machine Learning Model

Today we learn an extremely important lesson:

A model making predictions does not mean the model is good.

Learn

Study:

- Accuracy
- Confusion matrix
- Precision
- Recall
- True Positive
- True Negative
- False Positive
- False Negative

Don't worry about complicated mathematics yet. Understand what each measurement means.

Practical Work

Use yesterday's Iris model.

Program 1 – Accuracy

Calculate the model's accuracy.

Example:

Model Accuracy: 94%

Program 2 – Confusion Matrix

Create a confusion matrix.

Understand what it tells you.

For example:

	Predicted		
	A	B	C
Actual A	15	0	0
Actual B	0	13	2
Actual C	0	1	14

Program 3 – Classification Report

Use:

```
classification_report()
```

Display:

- Precision
- Recall
- F1-score

Questions to Answer

Write your answers in:

```
model_evaluation.txt
```

Question 1

If a model has:

```
95% accuracy
```

does that automatically mean it is a good model?

Question 2

Why might accuracy be misleading?

Question 3

What is the difference between precision and recall?

Question 4

For a system trying to detect a dangerous motor failure, which might be more important:

Missing a real fault or incorrectly warning about a healthy motor?

Explain your answer.

This question is deliberately connected to his future project.